

Sanmati Sudhakar Ukhalkar

AI&DS Engineer | Python

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PROFESSIONAL SUMMARY

AI&DS undergraduate with expertise in developing end-to-end machine learning systems. Proficient in Python with hands-on experience in Computer Vision, and Deep learning.

EDUCATION

- B.E. in Artificial Intelligence and Data Science** 2023 – 2026
SNJB's LSKBJ College of Engineering Nashik, Maharashtra
- Diploma in Computer Technology** 2020 – 2023
Shri H.J.B Polytechnic Nashik, Maharashtra Percentage: 70.69%
- 10th SSC Board** 2020
GSVM Elloral (SSC Board) Chh.Sambhajinagar, Maharashtra Percentage: 83.80%

PROJECTS

- CreatorPulse: Intelligent Social Media Management Engine** *React/TypeScript, Node.js/Express, FastAPI, OpenAI*
 - Developed an AI-powered workflow platform enabling content creators to generate tailored posts, automate scheduling, and track real-time engagement metrics.
 - Accelerated content creation pipelines and improved audience engagement by implementing AI-driven trend discovery and personalized creator-voice generation.
 - Architected a scalable tri-service system utilizing a React/TypeScript frontend, Node.js/Express backend, and Python/FastAPI AI engine powered by OpenAI.
 - Empowered digital entrepreneurs to effectively scale their online presence and foster stronger community connections through authentic, data-driven content strategies.
- Multi-Modal Recommendation Engine with Knowledge Graphs** *CNN, Transformers, Pandas, NumPy*
 - Designed and implemented A hybrid recommendation system that combines visual embeddings, user behavior patterns, and knowledge graphs to deliver personalized product recommendations across multiple modalities.
 - Built a system merging CNN-based image feature extraction, Transformer ensembles for sequential modeling, and Neo4j knowledge graph (50K+ nodes) to capture complex user-item-category relationships with graph traversal capabilities.
 - Implemented neural ranking with Transformer-based preference prediction, diversity re-ranking to avoid homogeneous suggestions, and real-time third-party API integration for dynamic content enrichment.
 - Achieved 85% precision@10 with 60% improvement over baseline collaborative filtering, optimized inference to sub-100ms latency, deployed caching layer for 70% throughput improvement, and successfully processes millions of user sessions in production.

TECHNICAL SKILLS

- Languages:** Python, SQL, JavaScript
- ML Frameworks:** TensorFlow, PyTorch
- Deep Learning:** CNNs, Transformers, GANs
- Computer Vision:** Object Detection, OpenCV, Image Processing
- NLP & LLMs:** Transformers, HuggingFace
- Data Tools:** Data Manipulation, NumPy, Pandas, Matplotlib, Seaborn,
- APIs & Backend:** FastAPI, API Integration, Docker

SOFT SKILLS

- Problem Solving, Collaborative Teamwork, Self-directed Learning, Attention to Detail